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AIR CLEANER

Abstract

Disclosed is an air cleaner. The air cleaner includes a case including a suction port disposed on a peripheral surface of the case, an air cleaning module including a blower fan disposed in the case and a filter configured to remove foreign matter from air introduced through the suction port, and a booster module disposed above the air cleaning module. The booster module including a head cover, a discharge port disposed inside the head cover, the discharge port including a ring-shaped inner grille having a central cavity, a display panel disposed in front of the inner grille, a front drive part connected to the display panel through the central cavity of the inner grille, and a first motor configured to move the display panel in a first direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Korean Patent Application No. 10-2024-0033178, filed on Mar. 8, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to an air cleaner, and more particularly to an air cleaner capable of discharging purified air in various directions.

2. Description of the Related Art

[0003] An air cleaner is a device that filters air in a certain space and discharges the filtered air, thereby reducing the concentration of dust or bacteria in the air in the corresponding space. The air cleaner filters out foreign matter by generating a flow of air in the corresponding space, and discharges air with the foreign matter removed therefrom.

[0004] In order to quickly purify air in the indoor space, a discharge port may be formed in an upper portion of an air cleaner, and filtered air discharged upwardly may be guided so as to flow in all directions.

[0005] Further, in the case in which a discharge port is formed in an upper portion of an air cleaner, a separate fan for regulation of the direction of air discharged may be disposed at an upper portion of the air cleaner in order to cause filtered air to flow a long distance.

[0006] Research on a multistage air cleaner in which blowing devices are stacked vertically is underway.

[0007] A related art document (Korean Patent Laid-Open Publication No. 10-2022-0007365) relates to a circulator for an air cleaner. Such a related art document discloses an air cleaner in which a circulator is disposed above blowing devices disposed in an upward-downward direction in order to cause purified air to flow a long distance.

[0008] However, the air cleaner disclosed in the above related art document has a limitation in that the circulator causes only a portion of purified air discharged from the upper blowing device to flow a long distance.

[0009] Further, the air cleaner disclosed in the above related art document has a problem of requiring an additional fan in order to send purified air away.

SUMMARY OF THE INVENTION

[0010] An aspect of the present disclosure is directed to providing an air cleaner capable of causing discharged purified air to circulate over a longer distance.

[0011] Another aspect of the present disclosure is directed to providing an air cleaner capable of sending a larger amount of purified air away.

[0012] A further aspect of the present disclosure is directed to providing an air cleaner capable of creating various types of airflows.

[0013] Still another aspect of the present disclosure is directed to providing an air cleaner capable of preventing discharged purified air from flowing back thereinto, thereby improving air cleaning performance.

[0014] Still another aspect of the present disclosure is directed to providing an air cleaner capable of preventing discharged purified air from flowing back thereinto without the necessity to add a structure such as a partition plate, thereby having a reduced size.

[0015] Still another aspect of the present disclosure is directed to providing an air cleaner capable

of forming an airflow that effectively removes allergens causing allergies.

[0016] Still another aspect of the present disclosure is directed to providing an air cleaner capable of improving suction capacity of air suctioned into the air cleaner.

[0017] The aspects of the present disclosure are not limited to the aspects mentioned above, and other aspects not mentioned herein will be clearly understood by those skilled in the art from the following description.

[0018] An air cleaner according to an embodiment of the present disclosure may deliver all of purified air discharged upwardly to a booster module, thereby sending a larger amount of purified air away.

[0019] An air cleaner according to an embodiment of the present disclosure may create various types of airflows without adding a separate fan by moving a display panel.

[0020] An air cleaner according to an embodiment of the present disclosure may create various types of airflows without adding a separate fan by rotating a booster module.

[0021] In an air cleaner according to an embodiment of the present disclosure, a booster module disposed above an air cleaning module includes a head cover, a ring-shaped inner grille having a central cavity formed therein and forming a discharge port inside the head cover, and a display panel disposed in front of the inner grille.

[0022] The booster module includes a motor configured to rotate the head cover.

[0023] The booster module includes a motor configured to move the display panel.

[0024] An inner flow guide is formed inside the head cover. The inner flow guide connects a front edge of the head cover to the inner grille. Air discharged through the inner grille may pass through a slit formed between the inner flow guide and the display panel.

[0025] An air cleaner according to an embodiment of the present disclosure includes a case having a suction port formed in a peripheral surface thereof, an air cleaning module including a blower fan disposed in the case and a filter configured to remove foreign matter from air introduced through the suction port, and a booster module disposed above the air cleaning module. The booster module includes a head cover, a ring-shaped inner grille having a central cavity formed therein and forming a discharge port inside the head cover, a display panel disposed in front of the inner grille, a front drive part connected to the display panel through the central cavity of the inner grille, and a first motor configured to move the display panel in a first direction, whereby various types of airflows may be created.

[0026] The booster module may further include a first gear connected to and rotated by the first motor and a second gear disposed on an inner surface of the front drive part and formed to be moved in cooperation with the first gear.

[0027] The booster module may further include a first booster motor support part configured to support the first motor and a booster neck configured to accommodate the first booster motor support part and support the inner grille from below. The booster neck may have an upper end formed so as to be inclined.

[0028] The display panel may be disposed parallel to the upper end of the booster neck.

[0029] The head cover may have a hemispherical shape including a first opening in which the display panel is disposed and a second opening in which the booster neck is disposed.

[0030] The first direction may be a direction perpendicular to a plurality of grilles interconnecting an inner wall of the inner grille and an outer wall of the inner grille.

[0031] The booster module may further include a second motor configured to rotate the head cover in a second direction, a third gear connected to and rotated by the second motor, and a fourth gear disposed on an inner surface of the head cover and formed to be moved in cooperation with the third gear. The second direction may be different from the first direction.

[0032] The air cleaner may further include an inner flow guide formed inside the head cover, and the inner flow guide may connect a front edge of the head cover to the inner grille.

[0033] Air discharged through the inner grille may pass through a slit formed between the inner

flow guide and the display panel.

[0034] The air cleaning module may include a first air cleaning module and a second air cleaning module disposed above the first air cleaning module.

[0035] Air purified by the second air cleaning module may be discharged through the inner grille.

[0036] The first air cleaning module may include a first discharge port formed in the bottom thereof, and air purified by the first air cleaning module may be discharged through the first discharge port formed in the bottom of the first air cleaning module.

[0037] The air cleaner may further include a bottom plate disposed below the first air cleaning module so as to be in contact with the floor, and an airflow guide may be disposed on the upper surface of the bottom plate so as to guide air discharged downwardly through the first discharge port in a lateral direction.

[0038] The airflow guide may include a curved portion, a flat portion extending laterally from the curved portion, and a boss protruding upward from an end portion of the flat portion.

[0039] A lower discharge grille including a plurality of grilles bent in an outward direction may be disposed inside the first discharge port.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0040] The above and other objects, features, and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0041] FIG. 1 is a view showing the external appearance of an air cleaner according to an embodiment of the present disclosure;

[0042] FIG. 2A is a cross-sectional view of the air cleaner according to the embodiment of the present disclosure;

[0043] FIG. 2B is an enlarged view of an upper end portion of the air cleaner shown in FIG. 2A;

[0044] FIGS. 3 and 4 are views for explaining operation of a booster module of the air cleaner according to the embodiment of the present disclosure;

[0045] FIGS. 5 and 6 are exploded views of the booster module of the air cleaner according to the embodiment of the present disclosure;

[0046] FIGS. 7 to 10 are cross-sectional views showing the operation modes of the booster module according to the embodiment of the present disclosure;

[0047] FIG. 11 is an assembled view of a first air cleaning module according to the embodiment of the present disclosure;

[0048] FIGS. 12 and 13 are exploded views of the first air cleaning module according to the embodiment of the present disclosure;

[0049] FIG. 14 is an enlarged view of a lower discharge airflow guide structure;

[0050] FIG. 15 is an assembled view of a control module according to the embodiment of the present disclosure;

[0051] FIGS. 16 and 17 are exploded views of the control module according to the embodiment of the present disclosure; and

[0052] FIGS. 18 and 19 are exploded views of a second air cleaning module according to the embodiment of the present disclosure.

DETAILED DESCRIPTION

[0053] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. The present disclosure may, however, be embodied in many different forms, and should not be construed as being limited to the embodiments set forth herein.

[0054] In the drawings, illustration of parts unrelated to the description is omitted to clearly and

briefly describe the present disclosure, and the same or extremely similar components are denoted by the same reference numerals throughout the specification.

[0055] As used herein, the terms with which the names of components are suffixed, “module” and “unit”, are assigned to facilitate preparation of this specification, and are not intended to suggest unique meanings or functions. Accordingly, the terms “module” and “unit” may be used interchangeably.

[0056] It will be understood that although the terms “first”, “second”, etc., may be used herein to describe various components, these components should not be limited by these terms. These terms are only used to distinguish one component from another component.

[0057] FIG. 1 is a view showing the external appearance of an air cleaner according to an embodiment of the present disclosure, FIG. 2A is a cross-sectional view of the air cleaner according to the embodiment of the present disclosure, and FIG. 2B is an enlarged view of an upper end portion of the air cleaner shown in FIG. 2A.

[0058] FIGS. 3 and 4 are views for explaining operation of a booster module of the air cleaner according to the embodiment of the present disclosure, and FIGS. 5 and 6 are exploded views of the booster module of the air cleaner according to the embodiment of the present disclosure.

[0059] First, referring to FIGS. 1, 2A, and 2B, a case **10** may have a suction port **10a** formed in the peripheral surface thereof to suction air from various directions. Air may be suctioned from all directions with respect to a central line passing through the internal center of the case **10** in an upward-downward direction. The suction port **10a** is provided in plural, and the plurality of suction ports **10a** is spaced apart from each other in the circumferential direction of the case **10**. The plurality of suction ports **10a** is evenly formed in the circumferential direction along the peripheral surface of the case **10** so that air is capable of being suctioned into the case **10** from any direction.

[0060] In this specification, the upward-downward direction or the vertical direction is defined as an axial direction. The axial direction may correspond to the central axis-direction of blower fans **140** and **240**, which will be described later, i.e., the motor shaft direction of the fans. The radial direction or the horizontal direction may be understood as a direction perpendicular to the axial direction. The circumferential direction may be understood as a circumferential direction of an imaginary circle that is formed when rotating about the axial direction with a distance in the radial direction as a rotational radius.

[0061] The air cleaner **1** includes air cleaning modules **100** and **200** that generate airflows. The air cleaning modules **100** and **200** include blower fans **140** and **240**, which are disposed in the case **10**, and filters **120** and **220**, which remove foreign matter from air introduced through the suction port **10a**, respectively.

[0062] The suction port **10a** allows the inside of the case **10** and the outside to communicate with each other. The suction port **10a** is provided in plural. The plurality of suction ports **10a** may be formed in the peripheral surface of the case **10**.

[0063] The plurality of suction ports **10a** is evenly formed in the circumferential direction along the outer circumferential surface of the case **10** so that air is capable of being suctioned into the case **10** from any direction.

[0064] The plurality of suction ports **10a** is formed in a stripe shape that is elongated in the upward-downward direction. Alternatively, the plurality of suction ports **10a** may be formed through perforation in a circular or elliptical shape.

[0065] As described above, because the case **10** is formed in a cylindrical shape and the plurality of suction ports **10a** is formed along the outer circumferential surface of the case **10**, the amount of air suctioned may increase.

[0066] The air introduced into the suction ports **10a** may pass through the filters **120** and **220**. Each of the filters **120** and **220** may be formed in a cylindrical shape and may have a filter surface that filters air.

[0067] The case **10** may have a cylindrical shape or a shape of a truncated cone (a cone with the top

cut off). A top cover **270** is disposed on the top of the case **10**. A booster module **300** is disposed on the top cover **270**.

[0068] Referring to FIGS. **1** to **6**, the booster module **300** may be coupled to the top cover **270** so as to seal the tops of the air cleaning modules **100** and **200**. The top cover **270** may be formed in a shape of a cylinder having a cavity formed in the central portion thereof. A portion of the booster module **300** may be inserted into the central cavity in the top cover **270** from above to below.

[0069] Referring to FIGS. **1** and **2A**, the air cleaning modules **100** and **200** include a first air cleaning module **100** and a second air cleaning module **200** that are disposed in the upward-downward direction.

[0070] For example, the second air cleaning module **200** may be disposed on the first air cleaning module **100**. Because the first air cleaning module **100** is disposed at a lower portion of the air cleaner **1**, the first air cleaning module **100** may be referred to as a “lower air cleaning module” or a “lower module”, and because the second air cleaning module **200** is disposed at an upper portion of the air cleaner **1**, the second air cleaning module **200** may be referred to as an “upper air cleaning module” or an “upper module”.

[0071] The first air cleaning module **100** and the second air cleaning module **200** include blower fans **140** and **240** and filters **120** and **220** configured to remove foreign matter from air introduced through the suction ports **10a**, respectively.

[0072] The first air cleaning module **100** includes a first blower fan **140** and a first filter **120** configured to remove foreign matter from air introduced through the suction ports. In addition, the first air cleaning module **100** further includes a first fan motor **130** configured to rotate the first blower fan **140**.

[0073] The second air cleaning module **200** includes a second blower fan **240** and a second filter **220** configured to remove foreign matter from air introduced through the suction ports. In addition, the second air cleaning module **200** further includes a second fan motor **230** configured to rotate the second blower fan **240**.

[0074] The air cleaning modules **100** and **200** include a discharge port **110a** formed in the bottoms thereof. The discharge port **110a** is formed below the first blower fan **140** and the first filter **120**. In the case in which the second air cleaning module **200** is disposed above the first air cleaning module **100**, the discharge port **110a** is formed in the bottom of the first air cleaning module **100**.

[0075] Air with foreign matter removed therefrom by the first filter **120** is discharged downwardly through the discharge port **110a**. Because the air purified in the first air cleaning module **100** is discharged through the discharge port **110a**, the discharge port **110a** may be referred to as a first discharge port **110a**.

[0076] The air cleaner **1** includes a discharge port **310a** formed in the booster module **300**. The discharge port **310a** is formed inside the booster module **300**. In addition, the discharge port **310a** is formed above the second blower fan **240** and the second filter **220**. That is, the discharge port **310a** is formed above the second air cleaning module **200**.

[0077] Air with foreign matter removed therefrom by the second filter **220** is discharged upwardly through the discharge port **310a**. Because the air purified in the second air cleaning module **200** is discharged through the discharge port **310a**, the discharge port **310a** may be referred to as a second discharge port **310a**.

[0078] In addition, a lower discharge grille **110** including a plurality of grilles is disposed at the bottom of the first air cleaning module **100** in order to guide a lower airflow. In addition, an inner grille **310** including a plurality of grilles is disposed inside the booster module **300** in order to guide an upper airflow.

[0079] The inner grille **310** may include an inner wall **311** and an outer wall **312**, which have a cylindrical shape. The plurality of grilles of the inner grille **310** may interconnect the inner wall **311** and the outer wall **312** and may be disposed in a radial form.

[0080] The lower discharge grille **110** may include a plurality of grilles having a concentric circle

structure. The plurality of grilles may be a plurality of circles having the same center and different radii. Each of the plurality of grilles may include a section bent in the radially outward direction. The lower discharge grille **110** may include a plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille **110** may be disposed inside the discharge port **110a** in order to deliver the lower discharge airflow in the lateral direction.

[0081] A bottom plate **170** is disposed below the first air cleaning module **100**. The bottom plate **170** is disposed in contact with the floor to support the air cleaning modules **100** and **200**.

[0082] An airflow guide **170a** (refer to FIG. 12) is disposed on the upper surface of the bottom plate **170** in order to guide air discharged downwardly through the first discharge port **110a** in the lateral direction. The bottom plate **170** may further include a base **170b** (refer to FIG. 12) that supports the airflow guide **170a**. The airflow guide **170a** may be formed on the upper surface of the base **170b**. In addition, the airflow guide **170a** and the base **170b** may be coupled to each other to constitute the bottom plate **170**.

[0083] According to the present disclosure, because the booster module **300** is integrated with the upper discharge passage of the air cleaner **1**, all the air flowing upward may pass through the booster module **300**, and accordingly, a larger amount of purified air may be sent away.

[0084] According to the present disclosure, although two air cleaning modules are stacked vertically, air is not discharged upwardly from the lower air cleaning module. Therefore, while suction performance is improved by forming the suction ports **10a** in the peripheral surface of the case **10**, it is possible to prevent air discharged from the lower air cleaning module from flowing back into the upper air cleaning module.

[0085] The 2-stage air cleaner of the related art has a problem in that purified air discharged from the lower air cleaning module flows upward and is suctioned into the upper air cleaning module, whereby air cleaning performance is reduced. In contrast, according to the present disclosure, purified air discharged from the lower air cleaning module flows downward, and thus introduction of the purified air into the upper air cleaning module is prevented, whereby air cleaning performance is improved.

[0086] Further, according to the present disclosure, the air cleaner **1** does not need to have a structure for preventing introduction of purified air discharged therefrom or configuration for discharge, the overall size of the product may be further reduced, and the freedom of design such as placement of internal parts may be improved.

[0087] For example, a control module **400** may be disposed at the center of the air cleaner **1**, thereby reducing the lengths of power lines and signal lines, simplifying a line connection structure, and reducing the overall height of the product. Further, the control module **400** may be disposed between the air cleaning modules **100** and **200** so that the air cleaning modules **100** and **200** are controlled by the single integrated control module **400**. Accordingly, each of the air cleaning modules **100** and **200** does not need to include an individual control module, and thus the overall height of the product may be reduced.

[0088] General allergen particles have a size of about 3 μm to about 100 μm , which is greater than the size of fine dust, and thus tend to settle on the floor. The lower airflow discharged downwardly from the first air cleaning module **100** causes allergen particles to float in the air and thus to be collected by the filters **120** and **220**. In this way, it is possible to effectively remove allergens causing allergies using the lower airflow discharged downwardly from the first air cleaning module **100**.

[0089] In addition, the air cleaner **1** may include a UVC LED (not shown), which is disposed in the case **10** and outputs ultraviolet in a wavelength band that has a sterilization effect on bacteria, mold, and microorganisms, and an ionizer (not shown), which is disposed in the case **10** and eliminates germs and mold using electricity supplied thereto.

[0090] The fan motors **130** and **230** may be accommodated in motor fastening parts **125** and **225**, respectively. The air cleaning modules **100** and **200** include filter mounting parts **160** and **260** in

which the filters **120** and **220** are mounted, respectively. The control module **400** is disposed between the filters **120** and **220**.

[0091] The annular first discharge port **110a** is formed in a lower portion of the first air cleaning module **100**. Air with foreign matter removed therefrom by the first filter **120** is discharged downwardly through the first discharge port **110a**.

[0092] The lower discharge grille **110** may include a plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille **110** may be disposed inside the first discharge port **110a** in order to deliver the lower discharge airflow in the lateral direction.

[0093] The first filter **120** may have a hollow cylindrical shape, and air may be introduced through the outer circumferential surface of the first filter **120**. While the air passes through the first filter **120**, impurities such as fine dust may be removed from the air.

[0094] Because the first filter **120** has a cylindrical shape, air is capable of being introduced into the first filter **120** from any direction. Therefore, the air filtering area may increase.

[0095] The first air cleaning module **100** includes a first fan housing **145** disposed below the first filter **120**, a first blower fan **140** rotatably disposed in the first fan housing **145**, and a first fan motor **130** configured to rotate the first blower fan **140**.

[0096] The first blower fan **140** suctions air in the axial direction and discharges the air downwardly in the radial direction. The first fan housing **145** may form a fan path that accommodates the first blower fan **140** and guides the air moved by the first blower fan **140** downwardly.

[0097] The second discharge port **310a** is formed in the booster module **300**. The inner grille **310** is disposed inside the second discharge port **310a** in order to change the direction of air flowing upward. For example, the second discharge port **310a** may be formed in a ring shape. The inner grille **310** may be formed in a radial shape.

[0098] Air with foreign matter removed therefrom by the second filter **220** is discharged upwardly through the second discharge port **310a**. The second air cleaning module **200** includes a second fan housing **245** disposed above the second filter **220**, a second blower fan **240** rotatably disposed in the second fan housing **245**, and a second fan motor **230** configured to rotate the second blower fan **240**.

[0099] The second blower fan **240** suctions air in the axial direction and discharges the air upwardly in the radial direction. The second fan housing **245** may form a fan path that accommodates the second blower fan **240** and guides the air moved by the second blower fan **240** upwardly.

[0100] The air moved by the second blower fan **240** flows to the booster module **300**.

[0101] The booster module **300** includes a head cover **320** forming the external appearance thereof and a display panel **390** configured to display various information such as an operation mode and indoor fine dust/harmful gas concentration.

[0102] The head cover **320** may have a hollow hemispherical shape. In this specification, the “hemispherical shape” may be a sphere with a portion cut off without being limited to half of a sphere.

[0103] The head cover **320** may form an inner path that accommodates parts and allows air introduced thereinto from the second air cleaning module **200** to move therealong. The head cover **320** may include a plurality of openings **320a** and **320b**.

[0104] The head cover **320** may include a first opening **320a** in which the display panel **390** is disposed and through which air is discharged and a second opening **320b** through which air is introduced thereinto from the second air cleaning module **200**.

[0105] The first opening **320a** may be formed so as to be inclined at a predetermined angle. The first opening **320a** may be disposed at a front portion of the head cover **320**. Here, the “front” may be a direction in which the display panel **390** that displays information to the user faces, and the “rear” may be a direction opposite the “front”.

[0106] The display panel **390** may be disposed in the first opening **320a** and may form the frontal appearance of the product. In addition, the display panel **390** and the head cover **320** may form a slit **351**, which is a space through which air passes.

[0107] The second opening **320b** may be formed in a lower portion of the head cover **320**.

Accordingly, air is introduced into the head cover **320** from the second air cleaning module **200** through the second opening **320b**. The air introduced into the head cover **320** through the second opening **320b** is discharged through the slit **351**, thereby forming the upper airflow.

[0108] The ring-shaped inner grille **310** having a central cavity formed therein and forming the second discharge port **310a** is disposed inside the head cover **320**. The display panel **390** is disposed in front of the inner grille **310**.

[0109] The air introduced into the head cover **320** through the second opening **320b** sequentially passes through the inner grille **310** and the slit **351**.

[0110] A front drive part **380** may be connected to the display panel **390** through a central cavity **313** in the inner grille **310**. The booster module **300** includes a first motor **330** configured to move the display panel **390** in a first direction.

[0111] A booster neck **360** may support the inner grille **310** from below. In addition, the booster neck **360** may support the front drive part **380**. The booster neck **360** may be located at the center of the second opening **320b**.

[0112] The booster neck **360** may be located above the second fan motor **230** so as not to interfere with the internal airflow. The booster neck **360** is coupled to the air cleaning modules **100** and **200**, which are main parts of the air cleaner **1**.

[0113] An upper portion of the booster neck **360** may be coupled to the booster module **300**. The upper portion of the booster neck **360** may be connected to an inner side of the inner grille **310**. The booster neck **360** is coupled to the inner side of the inner grille **310** in order not to block the airflow discharged from the inner grille **310**. That is, the booster neck **360** may be coupled to the inner wall **311** of the inner grille **310**.

[0114] A first booster motor support part **335** that supports the first motor **330** is accommodated inside the booster neck **360**. The first motor **330** may be coupled to the first booster motor support part **335**. Because the first motor **330** is also located inside the booster neck **360**, the first motor **330** does not block the inner path extending to the discharge port **310a** of the inner grille **310**.

[0115] The first motor **330** generates driving force for movement of the display panel **390**. For example, the first motor **330** may be a stepper motor, the rotational angle of which is easily controlled. In addition, a motor capable of rotating bidirectionally may be included in the first motor **330**.

[0116] The lower end of the booster neck **360** may be coupled to another mechanical structure of the second air cleaning module **200**. For example, the lower end of the booster neck **360** may be coupled to an upper side of the second fan housing **245**. Alternatively, the lower end of the booster neck **360** may be coupled to an upper side of the second motor fastening part **225**. In another embodiment, a separate coupling part (not shown) may be provided between the lower end of the booster neck **360** and the upper end of the second fan housing **245** or the upper end of the second motor fastening part **225**.

[0117] The front drive part **380** may be connected to the display panel **390**, and accordingly, at least a portion thereof may be moved along with movement of the display panel **390**.

[0118] The upper end of the booster neck **360** may be formed so as to be inclined. Accordingly, the inner grille may also be disposed so as to be inclined. In addition, the display panel **390** may be disposed parallel to the upper end of the booster neck **360**. Accordingly, it is possible to increase the visibility of information displayed on the display panel **390** and to send purified air, discharged through the slit **351** between the display panel **390** and the head cover **320**, away.

[0119] A display base **350** may support the display panel **390** from below, and may have a space defined therein so as to accommodate a printed circuit board. The display base **350** may be moved

together with the display panel **390**. Therefore, the movement of the display panel **390** may mean movement of the display base **350**.

[0120] The slit **351** through which air passes may be formed between the lower surface of the display base **350** and the head cover **320**. Accordingly, the display base **350** may guide air discharged through the second discharge port **310a** in the radially outward direction.

[0121] The first motor **330** may move the display panel **390** in the first direction, thereby changing at least one of the shape, size, or angle of the slit **351**. The form of the discharge airflow may be varied depending on movement of the display panel **390**.

[0122] Depending on the movement direction and movement distance of the display panel **390**, the size of a space between the display panel **390** and the head cover **320** may be varied, and the flowing direction and flowing distance of the air discharged from the second air cleaning module **200** may be varied.

[0123] The first direction may be a direction perpendicular to the plurality of grilles interconnecting the inner wall **311** and the outer wall **312** of the inner grille **310**. For example, the first direction may be a forward direction or a backward direction.

[0124] A first gear **331** is connected to the first motor **330**. The first gear **331** may be coupled to the motor shaft of the first motor **330**. The first gear **331** may be connected to and rotated by the first motor **330**.

[0125] In addition, a second gear **381** extending in the upward-downward direction is disposed on the inner surface of the front drive part **380**.

[0126] The first gear **331** is tooth-engaged with the second gear **381**. The second gear **381** may operate in cooperation with the first gear **331** and may extend in the forward direction. The second gear **381** is a guide rail that converts rotational motion into linear motion and includes a plurality of gear teeth. The first gear **331** also includes a plurality of gear teeth. The second gear **381** may be a type of rack gear including a plurality of gear teeth disposed in the longitudinal direction thereof. When the first motor **330** is driven, the first gear **331** rotates while being engaged with the second gear **381**.

[0127] As the first motor **330** is driven, the first gear **331** is rotated, and the second gear **381** is moved in cooperation therewith. If the first gear **331** is rotated in a predetermined direction by the first motor **330**, the second gear **381** of the front drive part **380** may be moved upward. Conversely, if the first gear **331** is rotated in the reverse direction by the first motor **330**, the second gear **381** of the front drive part **380** may be moved downward.

[0128] In some embodiments, the front drive part **380** may include a coupling portion (not shown), which is connected to the booster neck **360**, and a moving portion (not shown), on which the second gear **381** is mounted and which is connected to the coupling portion and the display base **350**. As the first motor **330** is driven, the moving portion may move the display panel **390** and the display base **350** while moving forward or backward.

[0129] FIG. **3** is a view illustrating a state in which the display panel **390** is moved forward, and FIG. **4** is a view illustrating a state in which the display panel **390** is moved backward.

[0130] The first motor **330** may move the display panel **390** forward and backward and may control the form of the airflow discharged from the booster module **300**. For example, when the display panel **390** is moved forward, air discharged through the slit **351** may move forward a long distance. When the display panel **390** is moved backward, air discharged through the slit **351** may move in the lateral direction.

[0131] According to the present disclosure, it is possible to create various types of airflows without adding a fan, and because the gear structure does not block the airflow path, air cleaning performance may be improved.

[0132] An inner flow guide **321** is formed inside the head cover **320**. The inner flow guide **321** may connect a front edge **322** of the head cover **320** to the inner grille **310**.

[0133] The inner surface of the inner flow guide **321** may be formed in a curved shape in order to

reduce flow resistance and change the flow direction of air discharged through the discharge port **310a** of the inner grille **310**.

[0134] The slit **351** is formed between the inner flow guide **350** and the display panel **390**. Air discharged through the discharge port **310a** of the inner grille **310** passes through the slit **351** and is discharged from the booster module **300**.

[0135] The booster module **300** may include a second motor **340** configured to rotate the head cover **320** in a second direction. The second motor **340** may be a stepper motor, the rotational angle of which is easily controlled. In addition, a motor capable of rotating bidirectionally may be included in the second motor **340**. The second motor **340** may be coupled to a second booster motor support part **345**.

[0136] A third gear **341** is connected to the second motor **340**. The third gear **341** may be coupled to the motor shaft of the second motor **340**. The third gear **341** may be connected to and rotated by the second motor **340**.

[0137] In addition, a fourth gear **325** is disposed on the inner surface of the head cover **320** so as to extend along the inner surface of the head cover **320**. The head cover **320** may include a curved portion **323** extending downward from the front edge **322** thereof. The fourth gear **325** may be disposed at the lower end of the inner surface of the curved portion **323**.

[0138] The third gear **341** is tooth-engaged with the fourth gear **325**. The fourth gear **325** may operate in cooperation with the third gear **341** and may extend in the forward direction. The fourth gear **325** is a guide rail that converts rotational motion into linear motion and includes a plurality of gear teeth. The third gear **341** also includes a plurality of gear teeth. The fourth gear **325** may be a type of rack gear including a plurality of gear teeth disposed in the longitudinal direction thereof. When the second motor **340** is driven, the third gear **341** rotates while being engaged with the fourth gear **325**.

[0139] As the second motor **340** is driven, the third gear **341** is rotated, and the fourth gear **325** is moved in cooperation therewith. If the third gear **341** is rotated in a predetermined direction by the second motor **340**, the fourth gear **325** of the head cover **320** may be moved upward. Conversely, if the third gear **341** is rotated in the reverse direction by the second motor **340**, the fourth gear **325** of the head cover **320** may be moved downward.

[0140] As the second motor **340** is driven, the head cover **320** may be rotated in the second direction. The second direction may be different from the first direction. Accordingly, it is possible to create a wider variety of airflows.

[0141] In general, the air cleaner of the related art is provided with an additional fan in order to send purified air away. Further, because the air cleaner discharge path and the clean booster path are separated from each other, only a portion of the purified air discharged from the air cleaner is sent away via the fan.

[0142] In contrast, according to the present disclosure, because the air cleaner discharge path and the clean booster path are integrated with each other, it is possible to send purified air away without a separate fan and to control the direction and form of the discharge airflow by moving the display panel **390**.

[0143] FIGS. 7 to 10 are cross-sectional views showing the operation modes of the booster module according to the embodiment of the present disclosure.

[0144] Referring to FIG. 7, as the first motor **330** is driven, the first gear **331** is rotated, and the second gear **381** is moved forward. The second gear **381** is disposed inside the front drive part **380**, and the display panel **390** and the display base **350** are connected to the front drive part **380**. Therefore, the display panel **390** is moved forward in accordance with the movement of the second gear **381**.

[0145] When the display panel **390** is moved forward, the slit **351** between the display panel **390** and the head cover **320** is widened sufficiently so as to allow air to travel forward. Therefore, when the display panel **390** is moved forward, a front airflow is formed.

[0146] Referring to FIG. 8, as the first motor **330** is driven, the first gear **331** is rotated, and the second gear **381** is moved backward. The display panel **390** is moved backward in accordance with the movement of the second gear **381**.

[0147] When the display panel **390** is moved backward, the slit **351** between the display panel **390** and the head cover **320** is narrowed. Therefore, when the display panel **390** is moved backward, a lateral airflow is formed along the narrowed slit **351**.

[0148] Referring to FIG. 9, as the second motor **340** is driven, the third gear **341** is rotated in the clockwise direction, and the fourth gear **325** is rotated in the clockwise direction when viewed from the side. The head cover **320** may be driven downward in accordance with the clockwise rotation of the fourth gear **325**.

[0149] Referring to FIG. 10, as the second motor **340** is driven, the third gear **341** is rotated in the counterclockwise direction, and the fourth gear **325** is rotated in the counterclockwise direction when viewed from the side. The head cover **320** may be driven upward in accordance with the counterclockwise rotation of the fourth gear **325**.

[0150] As shown in FIGS. 9 and 10, when the head cover **320** is driven upward and downward, the slits **351a** and **351b** are formed asymmetrically, thereby creating a wider variety of airflows.

[0151] FIG. 11 is an assembled view of the first air cleaning module according to the embodiment of the present disclosure, and FIGS. 12 and 13 are exploded views of the first air cleaning module according to the embodiment of the present disclosure.

[0152] Referring to FIGS. 11 to 13, the first discharge port **110a** that is open downward is formed in a lower portion of the first air cleaning module **100**. The lower discharge grille **110** that changes the direction of air flowing downward is disposed inside the first discharge port **110a**.

[0153] The annular first discharge port **110a** is formed in a lower portion of the first air cleaning module **100**. Air with foreign matter removed therefrom by the first filter **120** is discharged in the downward direction through the first discharge port **110a**.

[0154] The lower discharge grille **110** may include a plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille **110** may be disposed inside the first discharge port **110a** in order to deliver the lower discharge airflow in the lateral direction.

[0155] The first air cleaning module **100** includes a first filter mounting part **160** to which the first filter **120** is mounted. The first filter **120** may be removably mounted to the first filter mounting part **160**. The first filter **120** may have a hollow cylindrical shape, and air may be introduced through the outer circumferential surface of the first filter **120**. While the air passes through the first filter **120**, impurities such as fine dust may be removed from the air.

[0156] Because the first filter **120** has a cylindrical shape, air is capable of being introduced into the first filter **120** from any direction. Therefore, the air filtering area may increase.

[0157] The first filter mounting part **160** may be formed in a cylindrical shape corresponding to the shape of the first filter **120**. In the process of mounting the first filter **120**, the first filter **120** may be slidably fitted into the first filter mounting part **160**. Conversely, in the process of demounting the first filter **120**, the first filter **120** may be slidably withdrawn from the first filter mounting part **160**.

[0158] The first air cleaning module **100** includes a first fan housing **145** disposed below the first filter **120**, a first blower fan **140** rotatably disposed in the first fan housing **145**, and a first fan motor **130** configured to rotate the first blower fan **140**. In addition, the first air cleaning module **100** may further include a wire cover **115** that protects a wire connected to the first fan motor **130**.

[0159] The first blower fan **140** suctions air in the axial direction and discharges the air downwardly in the radial direction. The first fan housing **145** may form a fan path that accommodates the first blower fan **140** and guides the air moved by the first blower fan **140** downwardly.

[0160] The first fan motor **130** may be supported by the first motor fastening part **125**. The rotation shaft of the first fan motor **130** may extend upward from the first fan motor **130** and may pass through the bottom of the first motor fastening part **125** to be connected to the first blower fan **140**.

[0161] First magnets **180** are disposed on two opposite sides of the first fan housing **145**. The first magnets **180** may be coupled to metallic pieces of the case **10**. The positions of the magnets and the metallic pieces may be set to be opposite those exemplified above. For example, the metallic pieces may be disposed on two opposite sides of the first fan housing **145**, and the magnets may be disposed at the case **10**.

[0162] A plurality of support portions **150** extends upward from the first fan housing **145**. For example, the support portions **150** may be beams elongated in the longitudinal direction. A wire cover **155** may be disposed on at least one surface of each of the plurality of support portions **150** so as to cover the support portions **150**. The support portions **150** and the wire cover **155** may be spaced apart from each other to define a space in which wires or the like are disposed.

[0163] A steel net **165** is disposed between the first filter **120** and the first blower fan **140** in order to prevent body parts of the user or any other object from entering the space in which the first blower fan **140** is disposed.

[0164] The bottom plate **170** is disposed below the first air cleaning module **100**. The bottom plate **170** is disposed in contact with the floor to support the air cleaning modules **100** and **200**.

[0165] The airflow guide **170a** is disposed on the upper surface of the bottom plate **170** in order to guide air discharged downwardly through the first discharge port **110a** in the lateral direction. The bottom plate **170** may further include a base **170b** that supports the airflow guide **170a**.

[0166] FIG. **14** is an enlarged view of the lower discharge airflow guide structure.

[0167] Referring to FIGS. **11** to **14**, the first discharge port **110a** is formed in the lower end of the first air cleaning module **100**, and the lower discharge grille **110** is disposed inside the first discharge port **110a** in order to guide the lower discharge airflow.

[0168] The lower discharge grille **110** may include a plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille **110** may be disposed inside the first discharge port **110a** in order to deliver the lower discharge airflow in the lateral direction.

[0169] When the first blower fan **140** is rotated by the first motor **130**, the air purified by the first filter **120** is discharged through the first discharge port **110a** as the lower discharge airflow.

[0170] The airflow guide **170a** is disposed on the upper surface of the bottom plate **170** in order to guide air discharged downwardly through the first discharge port **110a** in the lateral direction.

[0171] General allergen particles, which have a size of about 3 μm to about 100 μm , are larger and heavier than indoor floating dust particles, which have a size of about 1 μm , and thus tend to settle on the floor. For example, mold particles have a size of about 10 μm , household dust allergen particles have a size of about 15 μm , and pollen particles have a size of about 50 μm .

[0172] Because virus or cigarette smoke particles have a very small size, the same tends to continuously float without settling on the floor. Indoor floating dust particles remain suspended in the air for a very long time. For example, it takes about 9 to 10 hours for indoor floating dust particles to settle from the ceiling to the floor of the room.

[0173] However, under the same conditions, mold particles, household dust allergen particles, and pollen particles float for a short time of several seconds to several minutes and then settle on the floor. Therefore, it is not easy for the air cleaner of the related art to capture allergen particles. In addition, the only way to remove the settled particles in the related art is to use a vacuum cleaner.

[0174] According to the present disclosure, the lower airflow discharged downwardly from the first air cleaning module **100** may cause allergen particles to float in the air, and the floating allergen particles may be captured by the filters **120** and **220**.

[0175] Meanwhile, according to an embodiment of the present disclosure, the speed of the lower discharge airflow may be controlled in response to the terminal velocity of allergen particles. The speed of the lower discharge airflow may be set to be higher than the terminal velocity of allergen particles.

[0176] The rotational speed of the first blower fan **140** may be controlled based on the terminal velocity of allergen particles. The rotational speed of the first blower fan **140** may correspond to the

speed of the lower discharge airflow set to be higher than the terminal velocity of allergen particles. A controller mounted on a main printed circuit board (PCB) **430** of the control module **400** may control the rotational speed of the first blower fan **140**.

[0177] The terminal velocity of allergen particles may be calculated using balance between gravity, buoyancy, and drag force acting on allergen particles. For example, the terminal velocity of allergen particles may be calculated using Stokes' terminal velocity formula.

[0178] In order to cause mold particles having a size of about 10 μm and household dust allergen particles having a size of about 15 μm to float in the air, the speed of the lower discharge airflow may be set to be higher than the terminal velocity of larger household dust allergen particles. For example, the speed of the lower discharge airflow may be set to 0.0038 m/s or higher.

[0179] The airflow guide **170a** includes a curved portion **173**, a flat portion **171** extending laterally from the curved portion **173**, and a boss **172** protruding upward from an end portion of the flat portion **171**.

[0180] The boss **172** may form an airflow that effectively causes allergen particles to float in the air. The boss **172** may reduce flow resistance at the end portion of the flat portion **171**. The boss **172** may prevent flow separation at the end portion of the flat portion **171**. The boss **172** may be formed in a ring shape along the edge of the airflow guide **170a**. The boss **172** may protrude from the flat portion **171** by a predetermined height **d1**, thereby preventing the occurrence of turbulence at the end portion of the flat portion **171** and causing ascending of the airflow. Accordingly, it is possible to effectively cause allergen particles settled on the floor to float in the air while preventing flow separation.

[0181] If the height **d1** of the boss **172** is too great, it may excessively increase the degree of ascending of the airflow, which is inappropriate for causing allergen particles settled on the floor to float in the air. Therefore, it is desirable that the height **d1** of the boss **172** be set to several millimeters (mm) or less, so long as the boss **172** is capable of preventing flow separation.

[0182] The curved portion **173** may be formed so as to surround a central pillar **175** of the bottom plate **170**. The curved portion **173**, which imparts a curvature to the airflow guide **170a**, may be formed so as to be round with a predetermined radius of curvature **R1**. The radius of curvature **R1** may be set so as to guide the lower discharge airflow to smoothly flow in the lateral direction.

[0183] The grille bending angle of the lower discharge grille **110** may be set to 20 degrees to 40 degrees so that the airflow rises at 20 degrees or greater in order to cause allergen particles on the floor to float in the air.

[0184] In addition, the radius of curvature **R1** may be set corresponding to the grille bending angle of the lower discharge grille **110**. For example, a predetermined angle **ag1** based on the vertical direction may be determined corresponding to the grille bending angle of the lower discharge grille **110**, and the radius of curvature **R1** may be set so as to implement the determined angle **ag1**. In more detail, the angle **ag1** may be set to be smaller than the grille bending angle of the lower discharge grille **110**.

[0185] The predetermined angle **ag1** may be an angle between the central pillar **175** and the curved portion **173**. In this case, the predetermined angle **ag1** may be smaller than the angle of the discharge airflow through the discharge port **110**. The angle of the discharge airflow may correspond to the grille arrangement angle and grille bending angle of the lower discharge grille **110**.

[0186] The curved portion **173** may guide the lower discharge airflow to smoothly flow in the lateral direction, thereby reducing flow resistance. According to the present disclosure, in order to deliver the lower discharge airflow in the lateral direction, the bottom plate **170** is imparted with a curvature, and the boss **172** is formed at an end portion of the flat portion **171**. Accordingly, a turbulent flow for causing allergen particles to float in the air may be created. That is, because the airflow guide **170a** is imparted with an inclination and a curvature and is provided with the boss for preventing flow separation, resistance to the lower discharge airflow may be minimized.

[0187] FIG. 15 is an assembled view of the control module according to the embodiment of the present disclosure, and FIGS. 16 and 17 are exploded views of the control module according to the embodiment of the present disclosure.

[0188] Referring to FIGS. 15 to 17, the control module 400 includes an upper cover 410 and a lower cover 420 that are coupled to each other to form the external appearance of the control module 400. The control module 400 includes a main PCB 430 on which circuits, such as a controller for control of overall operation of the air cleaner 1, are mounted, and the main PCB 430 is accommodated in an inner space defined by the upper cover 410 and the lower cover 420.

[0189] Each of the upper cover 410 and the lower cover 420 has a central cavity formed therein, and an upper filter pressing plate 440 and a lower filter pressing plate 450 are disposed in the central cavity of the upper cover 410 and the central cavity of the lower cover 420, respectively. Each of the upper filter pressing plate 440 and the lower filter pressing plate 450 may be connected to a plurality of springs 435.

[0190] The control module 400 may be located between the first air cleaning module 100 and the second air cleaning module 200. The control module 400 may be located above the first air cleaning module 100 and below the second air cleaning module 200.

[0191] Because the control module 400 is disposed between the first air cleaning module 100 and the second air cleaning module 200, the lengths of wires connecting the control module 400 to the first and second air cleaning modules 100 and 200 and variation in the lengths of the wires may be minimized, and the wiring structure may be simplified.

[0192] Further, because the filter pressing plates 440 and 450 are integrated with the upper and lower covers 410 and 420, vibration and noise may be further reduced, and manufacturing costs may be reduced.

[0193] When the first filter 120 is mounted to the first filter mounting part 160 of the first air cleaning module 100, the first filter 120 presses the lower filter pressing plate 450. When the second filter 220 is mounted to the second filter mounting part 260 of the second air cleaning module 200, the second filter 220 presses the upper filter pressing plate 440. Accordingly, the filters 120 and 220 may be more securely supported.

[0194] In addition, the control module 400 includes an upper UVC LED 481 and a lower UVC LED 482. The upper UVC LED 481 outputs UVC light upwardly toward the second air cleaning module 200 and the second filter 220. The lower UVC LED 482 outputs UVC light downwardly toward the first air cleaning module 100 and the first filter 120.

[0195] In general, ultraviolet light may be subdivided according to wavelength. For example, ultraviolet light may be classified into UVA having a wavelength of 320 nm to 400 nm, UVB having a wavelength of 280 nm to 320 nm, and UVC having a wavelength of 200 nm to 280 nm. In particular, UVC, which is short-wavelength ultraviolet light, has characteristics of destroying DNA of bacteria and causing a chemical reaction with a specific substance, and thus is effective for sterilization.

[0196] The control module 400 includes a mounting part 460 that supports the main PCB 430. Various sensors, such as an LED sensor 471, a laser sensor 472, and a gas sensor 473, may be mounted to the mounting part 460. Some of the sensors (e.g., the LED sensor 471) may include a cover 465.

[0197] The LED sensor 471 may emit infrared LED light and may detect light scattered by dust, thereby sensing the amount of dust. The laser sensor 472 may sense the amount of dust using laser light. The gas sensor 473 may sense the amount of gas in the air.

[0198] Although the LED sensor 471, the laser sensor 472, and the gas sensor 473 are illustrated in FIGS. 16 and 17, the disclosure is not limited thereto. The types of sensors provided and combinations thereof may be varied. In addition, one or more communication modules (not shown), such as a Wi-Fi module, may be provided inside the control module 400.

[0199] FIGS. 18 and 19 are exploded views of the second air cleaning module according to the

embodiment of the present disclosure.

[0200] Referring to FIGS. **18** and **19**, a hollow upper cover **270** is disposed at an upper portion of the second air cleaning module **200**. Air with foreign matter removed therefrom by the second filter **220** flows upward and enters the booster module **300** through the upper cover **270**.

[0201] In the second air cleaning module **200**, the second filter **220** is mounted to the second filter mounting part **260**. The above description of the first filter **120** and the first filter mounting part **160** may be similarly applied to the second filter **220** and the second filter mounting part **260**.

However, while the first filter **120** and the first filter mounting part **160** are located above the first blower fan **140** and the first fan housing **145**, the second filter **220** and the second filter mounting part **260** are located below the second blower fan **240** and the second fan housing **245**.

[0202] The second air cleaning module **200** includes a second fan housing **245** disposed above the second filter **220**, a second blower fan **240** rotatably disposed in the second fan housing **245**, and a second fan motor **230** configured to rotate the second blower fan **240**.

[0203] In addition, the second air cleaning module **200** may further include a wire cover **215** that protects a wire connected to the second fan motor **230**. The second air cleaning module **200** may further include a rotary bearing **290** that reduces frictional force generated during rotation.

[0204] The second blower fan **240** suctions air in the axial direction and discharges the air upwardly in the radial direction. The second fan housing **245** may form a fan path that accommodates the second blower fan **240** and guides the air moved by the second blower fan **240** upwardly.

[0205] The second fan motor **230** may be supported by the second motor fastening part **225**. The rotation shaft of the second fan motor **230** may extend downward from the second fan motor **230** and may pass through the top of the second motor fastening part **225** to be connected to the second blower fan **240**.

[0206] Second magnets **280** are disposed on two opposite sides of the second fan housing **245**. The second magnets **280** may be coupled to metallic pieces of the case **10**. The positions of the magnets and the metallic pieces may be set to be opposite those exemplified above. For example, the metallic pieces may be disposed on two opposite sides of the second fan housing **245**, and the magnets may be disposed at the case **10**.

[0207] A plurality of support portions **250** extends downward from the second fan housing **245**. For example, the support portions **250** may be beams elongated in the longitudinal direction. A wire cover **255** may be disposed on at least one surface of each of the plurality of support portions **250** so as to cover the support portions **250**. The support portions **250** and the wire cover **255** may be spaced apart from each other to define a space in which wires or the like are disposed.

[0208] A steel net **265** is disposed between the second filter **220** and the second blower fan **240** in order to prevent body parts of the user or any other object from entering the space in which the second blower fan **240** is disposed.

[0209] As is apparent from the above description, according to at least one of the embodiments of the present disclosure, a booster module, which includes a head cover, a ring-shaped inner grille having a central cavity formed therein and forming a discharge port inside the head cover, a display panel disposed in front of the inner grille, a front drive part connected to the display panel through the central cavity of the inner grille, and a first motor configured to move the display panel in a first direction is disposed above an air cleaning module, thereby creating various types of airflows without adding a fan.

[0210] According to at least one of the embodiments of the present disclosure, the booster module further includes a first gear connected to and rotated by the first motor and a second gear disposed on an inner surface of the front drive part and formed to be moved in cooperation with the first gear, whereby the gear structure does not block an airflow path, and thus air cleaning performance may be improved.

[0211] According to at least one of the embodiments of the present disclosure, the upper end of a

booster neck supporting the inner grille is formed so as to be inclined, and the display panel is disposed parallel to the upper end of the booster neck, whereby the visibility of information displayed on the display panel may be improved, and purified air may be sent away.

[0212] According to at least one of the embodiments of the present disclosure, the booster module further includes a second motor configured to rotate the head cover in a second direction, a third gear connected to and rotated by the second motor, and a fourth gear disposed on an inner surface of the head cover and formed to be moved in cooperation with the third gear, whereby a wider variety of airflows may be created.

[0213] According to at least one of the embodiments of the present disclosure, an inner flow guide is formed inside the head cover so as to connect a front edge of the head cover to the inner grille, thereby reducing internal flow resistance.

[0214] According to at least one of the embodiments of the present disclosure, the air cleaning module includes a first air cleaning module and a second air cleaning module disposed above the first air cleaning module, thereby improving air cleaning performance.

[0215] According to at least one of the embodiments of the present disclosure, air purified by the second air cleaning module is discharged through the inner grille, whereby all of the air purified by the second air cleaning module may be delivered to the booster module, and thus a larger amount of purified air may be sent away.

[0216] According to at least one of the embodiments of the present disclosure, a first discharge port is formed in the bottom of the first air cleaning module, and air purified by the first air cleaning module is discharged through the first discharge port formed in the bottom of the first air cleaning module, whereby a lower discharge airflow that is effective for removal of allergen particles may be created.

[0217] According to at least one of the embodiments of the present disclosure, an airflow guide is disposed on the upper surface of a bottom plate, which is disposed below the air cleaning module, in order to guide air discharged downwardly through the first discharge port in the lateral direction, whereby allergens causing allergies may be caused to float in the air and may be captured. Accordingly, allergens may be effectively removed.

[0218] According to at least one of the embodiments of the present disclosure, the airflow guide includes a boss protruding upward from an end portion thereof, thereby reducing flow resistance.

[0219] According to at least one of the embodiments of the present disclosure, a lower discharge grille including a plurality of grilles bent in the outward direction is disposed inside the first discharge port, thereby creating a lower discharge airflow that causes allergen particles to float in the air.

[0220] Although the present disclosure has been described with reference to specific embodiments shown in the drawings, it is apparent to those skilled in the art that the present disclosure is not limited to those embodiments and is embodied in many forms without departing from the scope of the present disclosure, which is described in the following claims. These modifications should not be individually understood from the technical spirit or scope of the present disclosure.

Claims

1. An air cleaner comprising: a case comprising a suction port disposed on a peripheral surface of the case; an air cleaning module comprising a blower fan disposed in the case and a filter configured to remove foreign matter from air introduced through the suction port; and a booster module disposed above the air cleaning module, the booster module comprising: a head cover; a discharge port disposed inside the head cover, the discharge port including a ring-shaped inner grille having a central cavity; a display panel disposed in front of the inner grille; a front drive part connected to the display panel through the central cavity of the inner grille; and a first motor configured to move the display panel in a first direction.

2. The air cleaner according to claim 1, wherein the booster module further comprises: a first gear configured to be rotated by the first motor; and a second gear disposed on an inner surface of the front drive part, the second gear being configured to be moved by the first gear.
3. The air cleaner according to claim 1, wherein the booster module further comprises: a booster motor support part configured to support the first motor; and a booster neck configured to accommodate the booster motor support part and support the inner grille.
4. The air cleaner according to claim 3, wherein an upper end of the booster neck is inclined with respect to a vertical direction.
5. The air cleaner according to claim 4, wherein the display panel is disposed to be parallel to the upper end of the booster neck.
6. The air cleaner according to claim 3, wherein the head cover has a curved shape and comprises a first opening configured to include the display panel, and a second opening configured to include the booster neck.
7. The air cleaner according to claim 1, wherein the inner grille comprises a plurality of grilles connecting an inner wall of the inner grille and an outer wall of the inner grille, and wherein the first direction is a direction perpendicular to a length of the plurality of grilles.
8. The air cleaner according to claim 1, wherein the booster module further comprises: a second motor configured to rotate the head cover in a second direction; a third gear configured to be rotated by the second motor; and a fourth gear disposed on an inner surface of the head cover, the fourth gear being configured to be moved by the third gear.
9. The air cleaner according to claim 8, wherein the second direction is different from the first direction.
10. The air cleaner according to claim 1, further comprising an inner flow guide disposed inside the head cover, the inner flow guide connecting a front edge of the head cover to the inner grille.
11. The air cleaner according to claim 10, wherein the inner grille is configured to discharge air through at least one slit disposed between the inner flow guide and the display panel.
12. The air cleaner according to claim 1, wherein the air cleaning module comprises: a first air cleaning module; and a second air cleaning module disposed above the first air cleaning module, and wherein the inner grille is configured to discharge air filtered by the second air cleaning module.
13. The air cleaner according to claim 12, wherein the first air cleaning module comprises a first discharge port disposed in a bottom of the first air cleaning module, the first discharge port being configured to discharge air filtered by the first air cleaning module.
14. The air cleaner according to claim 13, further comprising a bottom plate disposed below the first air cleaning module, the bottom plate being configured to be in contact with a floor, the bottom plate comprising an airflow guide disposed on an upper surface of the bottom plate, the airflow guide being configured to guide air discharged downwardly through the first discharge port in a lateral direction.
15. The air cleaner according to claim 14, wherein the airflow guide comprises: a curved portion; a flat portion extending laterally from the curved portion; and a boss protruding upward from an end portion of the flat portion.
16. The air cleaner according to claim 13, further comprising a lower discharge grille disposed inside the first discharge port, the lower discharge grille comprising a plurality of grilles bent in an outward direction.
17. An air cleaner comprising: a case comprising a suction port disposed on a peripheral surface of the case; an air cleaning module comprising a blower fan disposed in the case and a filter configured to remove foreign matter from air introduced through the suction port; and a booster module disposed above the air cleaning module, the booster module comprising: a head cover; a discharge port disposed inside the head cover, the discharge port including a ring-shaped inner grille having a central cavity; a display panel disposed in front of the inner grille; and a motor

configured to rotate the head cover.

18. The air cleaner according to claim 17, wherein the booster module further comprises: a front drive part connected to the display panel through the central cavity of the inner grille; and a motor configured to move the display panel through the front drive part.

19. The air cleaner according to claim 18, further comprising an inner flow guide disposed inside the head cover, the inner flow guide connecting a front edge of the head cover to the inner grille.

20. The air cleaner according to claim 19, wherein the inner grille is configured to discharge air through at least one slit disposed between the inner flow guide and the display panel.
